

- *Real-time mobile personalisation:* Promoting the experience
- *Fraud detection:* Detecting pattern changes
- *Product recommendations:* Customer personalisation
- *Learning management systems:* Decision-making programs
- *Dynamic pricing:* Flexible pricing based on a need or demand
- *Natural language processing (NLP):* Understanding and conversing with humans

To achieve the above business models, different types of learning algorithms are used. The broad categories of algorithms that are used based on approach, inputs/outputs, the type of problem to be solved, etc, are listed below.

Supervised learning: This is a type of algorithm in which the model is first trained on input and the corresponding output data so that it can predict the output for a new set of inputs. Examples are regression analysis, classification, etc.

Unsupervised learning: When the system detects structures/patterns in data that may not be very apparent to us, it's called unsupervised learning. Here the data is not labelled, but the system detects the groups by identifying the commonalities in the data. An example is finding groups or clusters.

Semi-supervised learning: This model falls between the supervised and unsupervised learning models. Unlabelled data, when used with a small amount of labelled data, is known to improve learning accuracy considerably.

Reinforcement learning: This is a training method based on rewarding desired behaviours and/or punishing undesired ones. As an example, a computer chess game keeps learning from us and surpasses our ability, learning new moves we have not yet used.

Industry adoption of Python in machine learning

The most common projects among ML implementers are information

processing, NLP, planning and exploring, machine vision, and handling and control.

It is estimated that the global AI market will grow from US\$ 7.27 billion in 2019, to reach over US\$ 35.4 billion by 2025, and that AI and ML investments will continue to grow at a CAGR of 42.8 per cent, in the period 2019-2025.

Machine learning implementation software

A ML framework is an interface, library or tool, which allows developers to build ML models easily and quickly, without getting into the practical details of the underlying algorithms.

It provides a clear, concise way for defining ML models using a collection of pre-built, optimised components. This not only democratises the development of ML algorithms but also speeds up the process. Some of the key features of a good ML framework are:

- Optimised for performance
- Developer friendly; the framework uses traditional ways of building models
- Is easy to understand and code on
- Is not completely a black box
- Provides parallelisation to distribute the computation process

Overall, an efficient ML framework reduces the complexity of machine learning, making it accessible to more developers.

Some of the most useful examples of software for ML are listed below.

1. **TensorFlow:** Available in Python, C++, Haskell, Java, Go, Rust and JS languages
2. **Keras:** A Python library for deep learning
3. **Microsoft Cognitive Toolkit:** Open sourced, written in C++ and can be accessed from Python and C++
4. **Caffe:** Python and C++ interfaces, written in C++

The languages popular for developing ML applications are: Python, Java, R and Scala.

Python: Python is the most used programming language for machine learning. Its popularity is due to the increased development of ML frameworks available for this language. It is powerful for preprocessing data and for working with data directly.

The ML Python libraries are:

- **Scikit-learn:** This is for data mining and analysis, which optimises Python's ML usability.
- **NumPy:** This is used for scientific calculations.
- **SciPy:** This is used for advanced computation.
- **Pybrain:** This is used for machine learning.
- **Pandas:** This offers developers high-performance structures and data.

Java: Java is widely used in enterprise programming. It is the second most preferred language used by data scientists and ML developers.

In terms of the ML applications in industry, Java tends to be used more than Python for network security, to control cyber-attacks and for fraud detection. The ML Java libraries are:

- **Deeplearning4j:** This is an open source and distributed deep learning library.
- **MALLET:** This allows for ML applications on text, including natural language processing, topic modelling, document classification and clustering.
- **Weka:** This is a collection of ML algorithms to use for data mining tasks.

R: This open source programming language is used mostly for statistical computing. R is an easy-to-use programming language mainly used for data analysis, statistical computing and visualisation in ML. The most widely used packages for ML in R are:

- **Caret**, which is for creating predictive models.
- **randomForest** is for classification and regression.